2 Theoretical background

2.1 Coordinate systems

Global as well as local coordinate systems are described in SOFiMSHC as cartesian right-handed system X-Y-Z. Rotations are applied in a mathematical positive sense. Within record SYST a global gravity direction can be specified at parameter GDIR. This global gravity or 'downward' direction affects the default orientation of loads, supports and other geometric attributes of structural items within the model if not specified differently at the respective location.

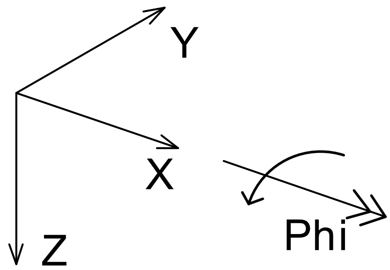

Figure 2.1: Coordinate system

If the observer is looking from the birds eye view he will believe to see a right or left handed 2D coordinate system depending on the orientation of the vertical axis . We use the designation of the "first" and the "second" horizontal axis in the counter clock wise orientation.

Each geometric or structural object in SOFiMSHC possesses a local orientation or a local coordinate system, which affects the direction of loads, cross-sections or support conditions:

- Points, for example, have a local coordinate system which defines primarily the local direction of supports and kinematic couplings. If no coordinate system is given explicitly the local z-direction defaults to the globally defined gravity direction or, if the point lies within a region or on a structural line, to the local coordinate system defined there.
- For structural lines, up to three different local coordinate systems can be identified. A first coordinate system is related to the underlying geometric curve and is primarily used to define the orientation of circular arcs or alignment axes. On the structural line, an independant coordinate system can be defined which sets the orientation of cross-sections and beam elements. A third coordinate system may be specified in order to set the local direction of supports, springs or kinematic couplings connected to a line. If one of the three mentioned directions is not explicitly set by the user, it defaults to the previously defined system. If no coordinate system is defined at all, the global gravity direction is used. The local coordinate system of a structural line is normally specified by the user by setting the direction of the local z-axis. As the local x-axis always points into the direction of the curve tangent, the local y-axis is defined automatically.
- Geometric surfaces and structural regions have a local coordinate system assigned which normally varies within the surface for curved shapes. The z-axis of the coordinate system always remains perpendicular on the surface. The coordinate system of a structural region defines, for example, the clock-order of outer boundary edges and the local orientation of the quadrilateral finite elements created on the surface.
- For volumes there might be a direction of orthotopic material properties, but there is no local coordinate system. However all surfaces describing the volume will have a unique interior

and exterior side. Thus a separating surface between two volumes will have a different orientation for the two cases.

2.2 Curves and alignment axes

Curves in SOFiMSHC are defined as parameter curves in three dimensional space. Parameter curves are basically defined by a local parameter s which runs along the curve from its start to its endpoint. A 'curve function' c(s) maps this local parameter s to global xyz-coordinates and therefore describes the curve in space when s is changed from smin to smax:

$$s \mapsto \vec{c}(s) = \begin{bmatrix} x(s) \\ y(s) \\ z(s) \end{bmatrix} \quad s = [s_{min}, s_{max}], \tag{2.1}$$

Apart from its shape other parameters might also be specified along a given curve as a function of s, like for example the orientation, the size or the shape of varying cross-sections. SOFiMSHC also allows to define so-called secondary lines, which are connected to a basis curve and whose distance is defined as a function of s.

2.2.1 Alignment axes

As a special type of curve SOFiMSHC allows to define alignment axes as primarily used in road and railway design. These curves typically consist of a sequence of straights and circular arcs with transition elements in between. In order to avoid sudden changes in curvature transition curves (or easement curves) are placed between sections with different radii providing a gradual change of curvature from one section to another. Depending on the characteristics of the curvature gradient different types of transition elements can be identified:

- Clothoid: Curvature varies linearly with distance s along the track

$$\kappa(s) = \frac{1}{r(s)} = \frac{s}{R \cdot L} = \frac{s}{A^2} \tag{2.2}$$

- Bloss-Curve: Curvature varies cubically with distance s

$$\kappa(s) = \frac{1}{r(s)} = \frac{3 \cdot s^2}{R \cdot L^2} - \frac{2 \cdot s^3}{R \cdot L^2} \tag{2.3}$$

- Sinusoidal transition curve:

$$\kappa(s) = \frac{1}{r(s)} = \frac{2\pi s - sin(\frac{2\pi s}{L})}{2\pi L R} \tag{2.4}$$

- Cosinusoidal transition curve:

$$\kappa(s) = \frac{1}{r(s)} = \frac{1 - cos(\frac{\pi s}{L})}{2R} \tag{2.5}$$

Above formulas apply for a transition curve of length L which starts from curvature=0 (straight axis) to a circular arc with radius R (curvature = 1/R). For transitions between sections with different radii (e.g. reversing clothoid, egg-shaped clothoid), they have to be modified accordingly. SOFiMSHC supports all variants.